Lab 01

Due January 27th 5:00 PM PST

STAT240 Spring 2026

2026-01-23

Instructions

For this homework, provide a rendered Quarto file in PDF format. You may render the Quarto file to HTML and then convert the HTML file to PDF using the print function in a web browser. This is exactly what you completed in Lab 0 last week. Indicate your student number on the quarto file (use the author option) and make a section for each problem.

Problem 1: Loading data

Part (a)

Write a code chunk in your file to do the following:

1. Assign your student number to a variable called `a`.
2. Print out the variable `a`.
3. Update the variable `a` by setting `a` to the value of `a` modulo 9 (`a %% 9`).
4. Update the variable `a` by adding 1 to `a`.
5. Print out the updated variable `a`.

(2 points)

Part (b)

The canvas folder for this lab also contains 10 datasets listed in the Excel document `datasets.xlsx`. Each dataset has two files:

1. One containing the data itself.
2. The other containing a description of the data.

Each of these 10 datasets has been assigned identification numbers (dataset IDs). The first column of `datasets.xlsx` is the dataset ID. You will analyze the dataset with ID given by your student number modulo 9 plus one (we will call this the dataset with ID `a`).

Write a code chunk (or multiple code chunks) in your file to do the following:

1. Load the data for the dataset with ID `a` using an appropriate call to `read.table` or `read.csv`.
2. Compute the mean of the `d`-th column of the dataset, and assign a variable to this mean. Here `d` is given by the column number specified in the column "Column" in `datasets.xlsx` for the dataset with ID `a`.
3. Compute the median of that same column (and assign a variable to it; and for the remaining 3 computations, assign the result of the computation to a variable).
4. Compute the standard deviation of that column.
5. Compute the minimum of that column.
6. Compute the maximum of that column.
7. Print out each of the 5 summary statistics computed above to 3 decimal places. Use the variables you assigned in the previous steps in the code used to print (don't just "hard code" the output by doing something like `print("3.142")`). Don't worry about leading line number indicators (like `[1]`) in the output.

Following this code chunk briefly write (in text, not in the code chunk):

- What you did
- Give the name of the column of the dataset you analysed (based on the associated information file)

Recall the rules for working directories when reading in files in a quarto document. Check the slides about this if you are unsure!

(8 points)